

Site Reliability Engineer, Observability - Take Home Project

Introduction

Congratulations on making it to the next part of our hiring process at Chainlink Labs. This is our Take Home Project, you're near the end! We appreciate the time invested in the process with us, and we are looking forward to receiving your submission.

During the exercise:

- If you have any questions, please reach out to your recruiter directly.
- Typically, the project can be completed within 4-6 hours.

Submitting the exercise:

- At Chainlink Labs, we present and communicate through Google documents. Please deliver your project in document/memo format unless otherwise asked.
- Submit your project directly to your recruiter and coordinator within 7 days of receiving this brief. Sharing as a PDF or .doc. Do not submit slides/decks.

Next steps:

- We will invite you to present your project to some of the interviewers you have already met. This is a 45-minute session; 30 minutes for presenting and 10-15 minutes for Q&A. You're welcome to invite questions and discussion throughout; we want to understand your thinking. We're not judging your presentation skills.

Task notes:

- You won't have all the context you need on how we work to complete this task to the highest levels of realism. Be creative and use assumptions about what we do. To help us understand your submission, share those assumptions with us.
 - It's fine if you're not sure about our current priorities; conjecture is OK.
 - We operate in a low-information environment, so you'll need to use your best judgment on what might be an interesting response to this task. We want to understand your ideas and approach, which is more important than trying to find a perfect outcome.
 - Please do not share or publish this exercise brief or your final work publicly.
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Project Prompt

Topic: Designing an End-to-End Observability Pipeline (Metrics, Logs, Traces)

Goal: We're looking to understand how you would design a complete local observability pipeline that ingests and stores metrics, logs, and traces from applications. Your solution should reflect strong judgment in system architecture, tooling, and operational best practices.

Task: Propose and document an approach for building an observability stack that operates entirely on your local machine.

You may select any architecture, tools, or strategies you believe are appropriate to ensure reliability, repeatability, and operability.

Requirements

- You define the architecture, tools, and observability signals.
- Use a local Kubernetes environment, if possible.
- Ensure data is persisted locally using Kubernetes PersistentVolumes.
- The system should be able to receive telemetry via OTLP, including: Metrics scraping (e.g., Prometheus pull model)
- Define three logical pipelines: one each for metrics, logs, and traces.
- Include dataflow diagrams (context + sequence) showing how telemetry flows through your system.
- Demonstrate basic internal operability signals of your choice (e.g., health checks, alerting, dashboards, service logs).

Optional (Bonus):

You're welcome to include a basic proof-of-concept setup, with example configuration or sample apps to demonstrate your design. This is not required, but appreciated.